Round 3: arity scaling of the tree (m = 5, 6) and the glued arm G

2026-11-03/04 · follow-up to `../round2/README.md` · predictions pre-registered in `.resume/RESUME.md`, commit `28fabab36d`, before any round-3 run · instance generator `../inst.sing` ⟶ `../instances_r3.txt` · generator `../gen.sing` (`INSTFILE`, `CELLS`) · input hashes `SYSTEMS.sha256` · logs `logs/` · records `rows.tsv` · graph `excess.svg` · PDF `report.pdf`

Class: stage diagnostic of the solver axis. No pipeline is priced and no leaderboard row changes. Engine: msolve 0.6.5 (F4, grevlex, 1 thread, GB only), p = 786433, curve as in `../instances.txt`, one planted target per cell.

Result

prediction	outcome
P1: D_S = 4 at m = 5, 6	**confirmed**: 4 in all four cells (m = 5, t = 4, 5; m = 6, t = 5, 6)
P2 (kill test): E(5) ≤ 5, E(6) ≤ 6, where E = D_T − D_S	**undecided**: E(5, t=4) ≥ 5 and E(6, t=5) ≥ 6. Both lower bounds sit exactly on the registered limits, and neither run finished on this 15 GB host
P3: G at m = 4, t = 4 has D in [7, 8]	**falsified**: G's input equation has total degree 16, and F4 reached degree 9 before the run was stopped. G is dominated by T (D = 7) and F (D = 8)

1. **The cherry floor holds up to m = 6.** With every internal partial sum fixed, D = 4 and every solve takes under a second (widest matrix ≤ 1,927 columns). The membership towers and the pair encodings still add no degree.
2. **Coupling excess E = D_T − 4.** At t = 4–5 it is 0, 2, 3 at m = 2, 3, 4 (measured) and at least 5 and at least 6 at m = 5, 6 (lower bounds). Averaged from m = 2 to 6, the lower bounds imply a slope of at least 1.5 per added summand, which would still be D = O(m), condition 1 of THEORY.md §3. But a lower bound cannot confirm a slope. The decisive cells need more than 13.5 GiB:
   - m = 5, t = 4 was cut off at 2,400 s while reducing degree 9 on a 572,048 × 999,374 matrix.
   - m = 6, t = 5 selected degree 10, then hit the 13.5 GiB cap at 1,304 s.
3. **Size, not only degree.** Even if E is linear, the matrices grow fast: $2.6 \cdot 10^4$ columns at m = 4, t = 4, and at least 10^6 at m = 5, t = 4. The cost law binom(N + D, D)^ω with N ≈ m·t and D ≈ 4 + 1.5·(m − 2) is what an L(1/2) claim would have to beat. These cells do not test that.
4. **Eliminating the separators backfires.** G = Res_X1(·, Res_X2(·, ·)) is S_5 written in the two pairs' (s, q). In (s, q) its standard total degree is 16 (degree 8 in each of s_0 and q_0). That is twice F's degree in $e_1..e_4$, because q = x·x′ counts as degree 1 while carrying x-degree 2.
   - F4 selected degree 16 (the input itself) and was in degree 9 at 83 s, when I stopped it to free cores for the kill-test cells. The t = 5 run was withdrawn for the same reason; its log records `Input file not found`.
   - A weighted grading (deg q = 2) would put G and F on equal footing. msolve does not take weights, so that comparison is not made here.

Reading. At these sizes the lever named in round 2, encoding the separators better, is not a resultant elimination: it moves the coupling into one high-degree equation, and F4 pays for it. The tree stays the only measured encoding with bounded input degree. Its excess is consistent with linear growth up to m = 6, but only as lower bounds. Whether its slope is ≤ 1 per summand (P2) needs a host with at least 64 GB.

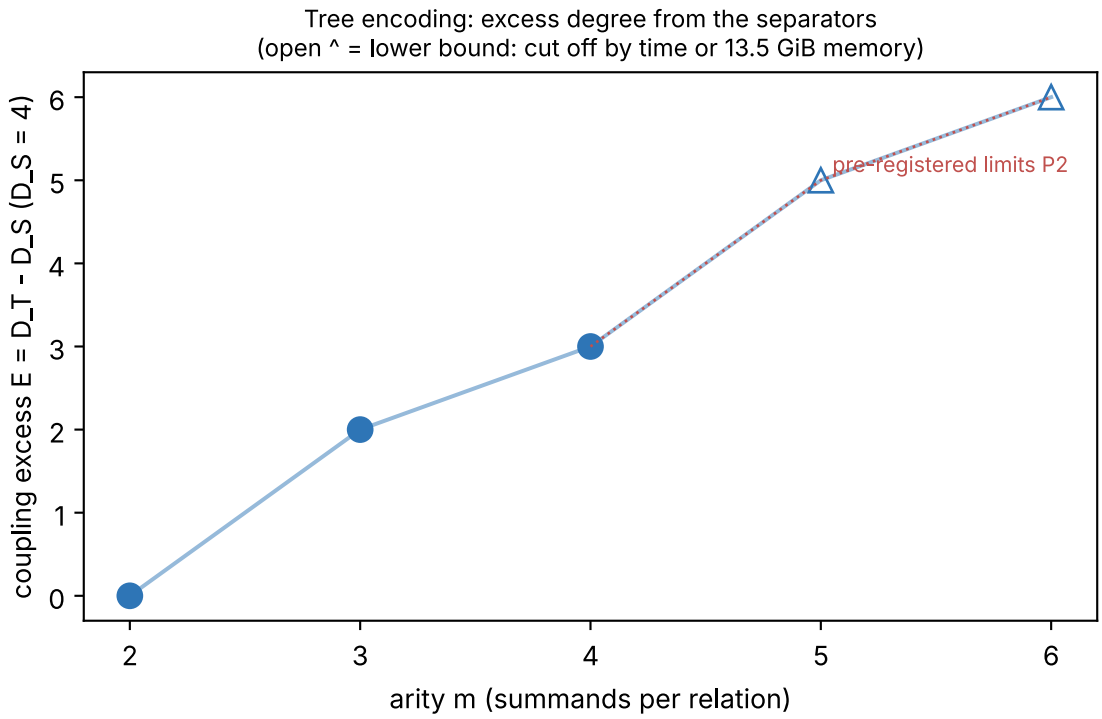


Table (recorded; `rows.tsv`)

cell	arm	N	D	widest matrix	wall s	exit	log
m5 t4	S	23	4	307	0.35	0	main
m5 t5	S	28	4	699	0.38	0	main
m6 t5	S	34	4	1,048	0.11	0	main
m6 t6	S	40	4	1,927	0.64	0	main
m5 t4	T	23	≥ 9	≥ 999,374 (round in progress)	2,400	124 timeout, 13.5 GiB	main
m6 t5	T	34	≥ 10 (selected)	≥ 734,728	1,304	139 memory, 13.5 GiB	main
m5 t4	T	23	≥ 9	≥ 654,028	620	139 memory, 6 GiB	memcap6g
m5 t5	T	28	≥ 9 (selected)	≥ 124,757	528	139 memory, 6 GiB	memcap6g
m6 t5	T	34	≥ 9	≥ 555,349	280	139 memory, 6 GiB	memcap6g
m6 t6	T	40	≥ 9	≥ 308,136	793	139 memory, 6 GiB	memcap6g
m4 t4	G	16	≥ 16 selected (input); working degree 9	≥ 72,406	83	143 stopped by operator	stopped

`parse.sh` counts only completed rounds. The m6 t5 degree-10 selection appears in the log's last, incomplete row (`10 95116 1780719`, then `exit=139`).

Deviations from the pre-registration

- m = 6 was registered at t = 4 and 5. At t = 4 the base has only 5 usable abscissas, too few for 6 distinct summands, so m = 6 ran at t = 5 and 6.
- The registered memory cap was 6 GiB with 2 jobs. All four T runs hit it, and the two deciding cells were rerun one at a time at 13.5 GiB.
- G was stopped before its t = 4 run finished, and its t = 5 run was withdrawn. Both decisions were made after seeing the input degree (16).

Checks

- Every round-3 system passed the planted check inside `gen.sing`. The instances come from `inst.sing`, natively and deterministically (LCG seed `0x7EE51103`). Internal consistency holds: cherry sum 1 equals caterpillar sum 1, and Y equals the 4-point partial sum.
- `gen.sing` still writes the round-1 systems byte-identical (60/60), and the round-2 spot checks are identical too (3/3).
- S's quotient dimension is 1 in every cell: fixed separators leave one solution.

Next

The remaining question is P2's slope at m = 5, 6. It needs a host with at least 64 GB, run through isolab-eks under the repository rule. The two cells, with their inputs, are pinned in `SYSTEMS.sha256`.